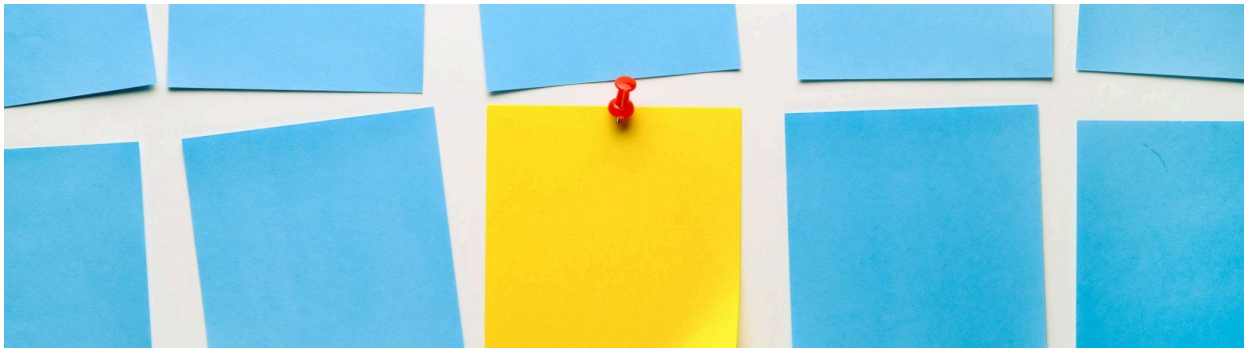


Heart Disease Prediction

Project PACE Document

Status Completed

Timing Jan 5, 2026 to Jan 19, 2026

Owners

Overview

Willber Escalante - AGE Analytics

Plan

Business Problem: Our Stakeholders are looking for an early-detection tool to help predict potential heart disease from routinely collected patient information

Stakeholders:

- *Primary*: AGE Analytics Team
- Secondary: Hypothetical Clinic/Clinical Stakeholders

Business Question: “Given the patient's vitals and test results, how likely is it to be able to determine whether or not they have heart disease?”

Goals: Perform Exploratory Data Analysis and create a classification machine learning model.

- We emphasize Recall because it's more costly to miss a positive diagnosis compared to a false positive. Follow up tests will mitigate the false positive diagnosis:
    - Aim for a Recall score of > or = 0.80
    - Aim for a ROC-AUC score of 0.75 - 0.85

Constraints & risks: Obtained a small publicly available dataset to use for predictive modeling. Additionally, this project does not perform clinical validation or model deployment. It is just a proof of concept using public data.

Analyze

Data sources:

- Kaggle - [Heart Disease Prediction Dataset] (https://www.kaggle.com/datasets/nalisha/heart-disease-prediction-dataset/data)

Key variables:

- cat_cols = ['sex', 'chest_pain_type', 'fbs_over_120', 'ekg_results', 'exercise_angina', 'slope_of_st', 'number_of_vessels_fluro', 'thallium']

- num_cols = ['age', 'bp', 'cholesterol', 'max_hr', 'st_depression']

- target_col = 'heart_disease_flag'

EDA focus areas:

- Cleaning and formatting data
  Identifying Patterns in Target Variable and Feature Variables in Distributions at a univariate and bivariate level

- Calculate Statistics

- Create visuals to understand relationships between features and target variable

Data Quality Issues:

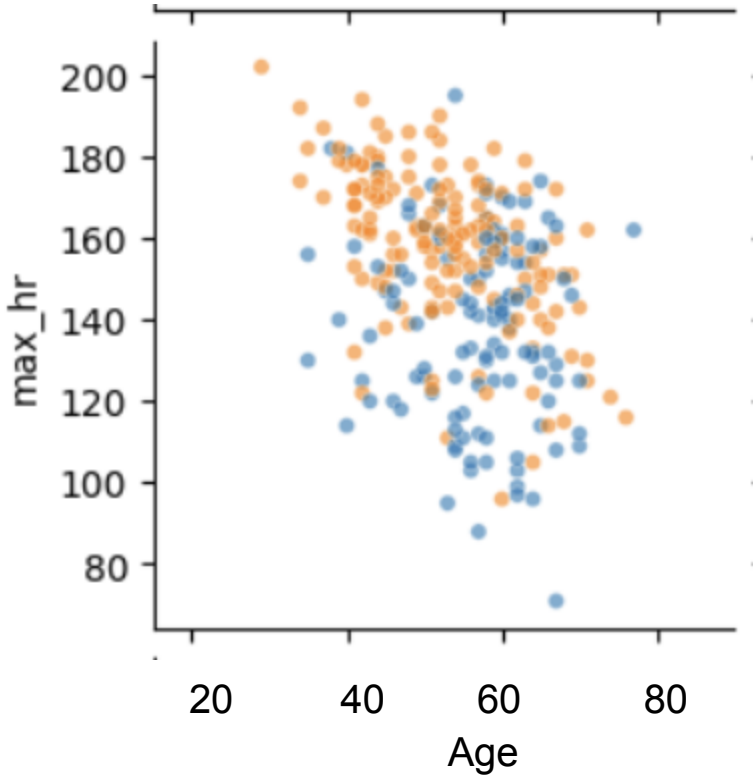
- Small Sample Size
- Kaggle Dataset

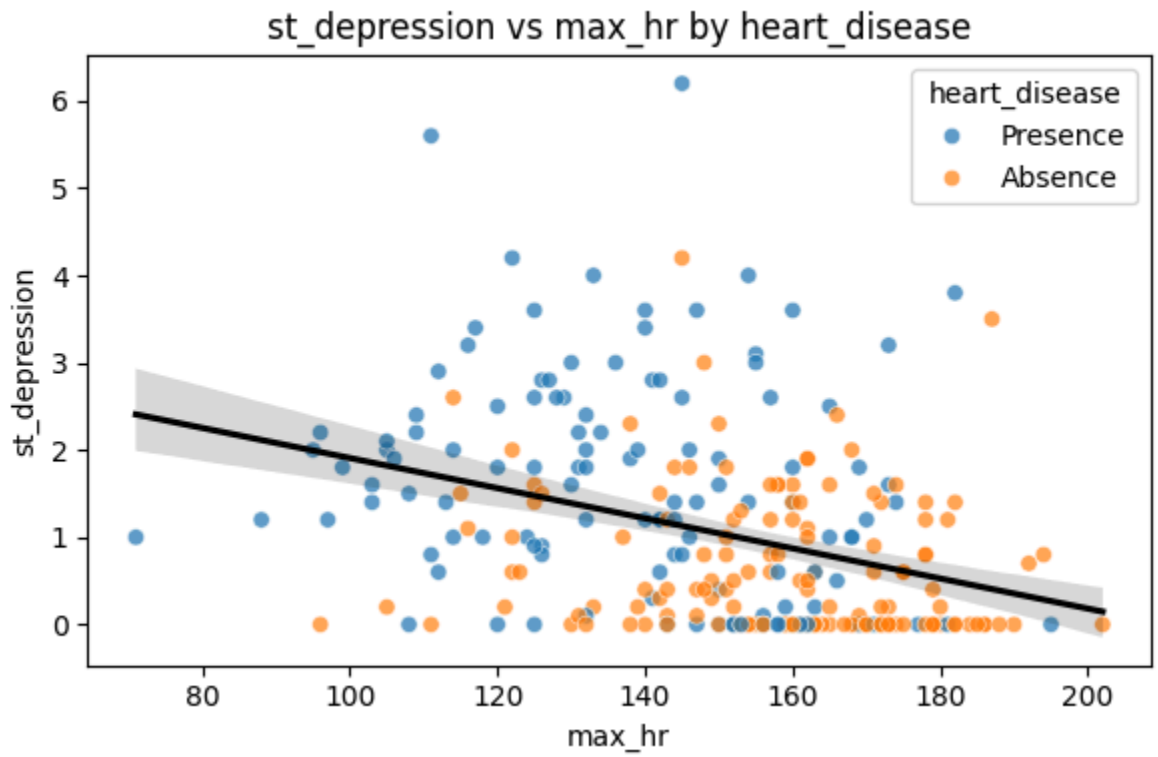
Insights:

- Age and Maximum Heart Rate are moderately negatively correlated
  - Patients older in age were associated with a lower maximum heart rate

- ST Depression and Maximum Heart Rate also show a moderately negative correlation. Higher ST Depression values were associated with lower maximum heart rate

- A Flat/Horizontal Slope of ST was associated with a higher rate of positive heart disease flag


max_hr
200
180
160
140
120
100
80
20
40
60
80
Age


st_depression vs max_hr by heart_disease
heart_disease
Presence
Absence
st_depression
0
1
2
3
4
5
6
80
100
120
140
160
180
200
max_hr

Construct

Modeling Approach: Begin with a linear statistical model (Logistic Regression) and compare it with a 2nd model (Random Forest). Tune hyperparameters if applicable.

Features and Transformation:

- Mapped the Target Variable as a binary option
- Create lists for categorical, numerical and target variables
- Build pipelines and preprocessors

Validation Strategy (Splits, etc):

- Split data into training, validation and test sets
- Instantiate and Fit models
- Evaluate the validation and test sets for each model

Model Choice:

- Two models were trained: Logistic Regression and Random Forest, using identical preprocessing pipelines
- Recall, F1 and ROC-AUC were prioritized to minimize missed cases
- Both models performed similarly with the validation set across the board. (Recall = 0.89 and ROC-AUC = 0.93 - 0.94)
- Logistic Regression Classifier was chosen due to its performance
    - Recall: 1.00
    - F1: 0.92
    - ROC-AUC: 0.99
    - These results indicate the model identifies most at-risk patients (high recall) while keeping false positives at an acceptable level (good precision/F1).

<u>Execute</u>

Deliverables:

- Cleaned Dataset
- Created Modeling Notebook/Script
- Model Performance Report
- PACE Document
- Slide Deck and PDF Executive Summary

Decisions enabled:

- AGE can demonstrate a proof of concept early detection model and estimate how model performance will affect screening decisions and resource allocation

Next steps / iteration plan: Validate model on larger/real clinical data, assess fairness across sex/age groups, and collaborate with clinicians before any deployment.

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Roles

Role	Name
Owners	Willber Escalante
Approvers	Willber Escalante
Contributors	Willber Escalante

Milestones

Date	Milestone	Description	Expected outcome
Jan 12, 2026	Milestone 1	Perform Initial EDA	• Clean and Format Data, Discover Data Distributions, Gather Insights on Feature on Feature Comparison
Jan 16, 2026	Milestone 2	Feature Engineering and Pre-Modeling	• Transform/Prepare data to create preprocessing pipelines and • Split data into Train/Validate/Test sets for modeling.
Jan 19, 2026	Milestone 3	Modeling, Model Comparison and Chosen Model Tuning	• Instantiate Classifiers, Calculate and Evaluate Metrics (Logistic Regression and Random Forest) • Tune hyperparameters to increase model efficiency.

Resources

Weekly report 🗋 File

Project brief 🗋 File

Brainstorming session 🗋 File

Research 🗋 File

Marketing assets 🗋 File